DOCUMENT 04

AI-BFSI Internship Project Documentation

Customer Journey Map

A successful software solution is defined not only by its features but also by the experience it provides throughout the user's interaction. This document illustrates the end-to-end customer journey for the Subscription Waste Detector, identifying how users interact with the system from uploading expense data to receiving AI-powered subscription recommendations.

Purpose of this Document

The purpose of this document is to analyse the complete user journey, identify interaction points, understand user objectives at each stage and demonstrate how the current implementation supports those interactions. The journey serves as a reference for improving usability, feature alignment and overall user experience.

Executive Summary

The Customer Journey Map describes the sequence of interactions that a user performs while using the Subscription Waste Detector. Unlike the previous Empathy Map, which focused on user thoughts and emotions, this document focuses on the actual workflow followed inside the application. Each stage highlights the user's objective, the system response and the value delivered by the implemented features.

The current implementation begins when a user provides expense data through CSV upload, Excel upload or manual expense entry. The uploaded information is processed using Python and Pandas to identify recurring subscription payments. The processed data is then presented through an interactive dashboard containing subscription summaries, spending visualisations, renewal reminders and AI-powered recommendations generated using the Groq API. The journey concludes when the user reviews these insights and makes informed decisions regarding recurring subscriptions.

Objectives

- Understand the complete customer journey.
- Identify every user interaction with the system.
- Highlight implemented features at each stage.
- Recognise opportunities for a better user experience.
- Support user-centred software improvement.

Journey Scope

This document covers the complete interaction flow of the current application, beginning with expense data submission and ending with the review of AI-generated recommendations. Features that are not part of the current implementation, such as authentication, cloud storage, database integration and bank connectivity, are intentionally excluded.

Document Outcome

By the end of this document, stakeholders will understand how users progress through the application, what value is delivered at each stage and how the implemented features collectively improve subscription visibility and financial awareness.

1. Customer Journey Overview

The customer journey illustrates the end-to-end interaction between the user and the Subscription Waste Detector. From uploading expense data to receiving AI-powered recommendations, every stage shown below represents functionality available in the current implementation.

End-to-End Customer Journey

User	Expense Input	Processing	Detection	Dashboard	AI Insights	Decision
👤	📂	⚙️	🔍	📊	🤖	✅
Starts Analysis	CSV Excel Manual	Python Pandas	Recurring Subscriptions	Charts Summary	Groq AI Suggestions	Review Take Action

Journey Summary

The application converts raw expense records into meaningful subscription insights through automated processing, recurring payment detection, interactive dashboards and AI-generated recommendations. Each stage contributes towards improving subscription awareness and supporting informed financial decisions.

2. Journey Stage Analysis

Each stage of the customer journey has a specific objective. The table below explains how the user interacts with the system and how the current implementation responds to those interactions.

Stage	User Goal	Journey Details
Data Input	Provide expense information.	**User Action:** Upload CSV, Excel or manually enter expenses. **System Response:** Accepts uploaded expense data for processing.
Processing	Organise uploaded data.	**User Action:** Wait while records are processed. **System Response:** Python and Pandas analyse uploaded expense data.
Detection	Identify recurring subscriptions.	**User Action:** Review detected subscriptions. **System Response:** Recurring transactions are identified and categorised.
Dashboard	Understand subscription spending.	**User Action:** View charts, summaries and subscription lists. **System Response:** Interactive Plotly visualisations present spending insights.
AI Assistance	Receive optimisation suggestions.	**User Action:** Review AI-generated recommendations. **System Response:** Groq AI generates personalised cost-saving suggestions.
Decision	Take informed financial action.	**User Action:** Evaluate subscriptions for future action. **System Response:** Consolidated insights support informed subscription decisions.

3. Pain Points and System Opportunities

The customer journey highlights several challenges encountered while managing recurring subscriptions. The Subscription Waste Detector addresses these challenges through features included in the current implementation, improving financial visibility and simplifying subscription management.

Pain Point 01

Scattered Expense Records

Subscription payments are distributed across bank statements, UPI transactions and digital payment histories, making them difficult to track collectively.

System Response

Centralized Subscription View

The application identifies recurring subscriptions and presents them in a single consolidated dashboard for easier review.

Pain Point 02

Limited Spending Awareness

Users often know individual transaction amounts but lack a clear view of their overall monthly subscription expenditure.

System Response

Visual Expense Analytics

Interactive charts and monthly summaries provide a quick understanding of recurring subscription spending.

Pain Point 03

Missed Renewal Dates

Automatic renewals frequently occur before users remember to review their subscriptions.

System Response

Renewal Reminders

The application displays upcoming subscription renewal information to improve financial awareness before recurring payments occur.

Pain Point 04

Uncertain Cost Optimization

Users may recognise recurring expenses but remain unsure which subscriptions should be reviewed or cancelled.

System Response

AI Recommendations

The Groq API analyses detected subscriptions and generates practical cost-saving recommendations for the user.

Journey Insight

Every major pain point identified during the customer journey is addressed by one or more implemented system features. Rather than adding complex functionality, the solution focuses on improving financial visibility and simplifying recurring subscription management.

4. Journey Mapping to Current Implementation

The following matrix demonstrates how each stage of the customer journey is supported by the existing implementation of the Subscription Waste Detector. Only features currently available within the application are included.

Journey Stage	Implemented Features	User Benefit
Expense Submission	CSV Upload Excel Upload Manual Expense Entry	Flexible methods for providing expense data.
Data Processing	Python Processing Pandas Analysis	Efficient organisation and preparation of uploaded data.
Subscription Detection	Recurring subscription identification Subscription categorisation	Automatic discovery of recurring payments.
Dashboard	Subscription summary Monthly spending Interactive Plotly charts	Improved financial visibility through visual analysis.
Decision Support	Renewal reminders Groq AI recommendations	Supports informed subscription review and optimisation.

Implementation Boundary

This customer journey reflects the current implementation only. Features such as authentication, cloud storage, bank API integration, automated notifications and database-backed user accounts are outside the present project scope and are considered future enhancements.

5. Conclusion

The Customer Journey Map demonstrates how the Subscription Waste Detector guides users from raw expense information to meaningful subscription insights through a structured and intuitive workflow. Each interaction has been designed to reduce manual effort while improving awareness of recurring financial commitments.

The journey validates that the implemented functionality aligns with the identified user needs. Expense data collection, subscription detection, interactive visualisation and AI-powered recommendations work together to create a straightforward user experience without unnecessary complexity.

Key Takeaways

- User interactions follow a simple sequential workflow.
- Every stage contributes measurable value to the overall experience.
- The implemented features directly address the primary user pain points.
- The application improves subscription awareness through automation and visual analysis.

Transition to Document 05

The customer journey establishes how users interact with the system. Building upon these interactions, the next document, **Requirements Gathering and Analysis**, formalises user expectations into structured software requirements that define the functional and non-functional capabilities of the Subscription Waste Detector.